

STANDARD OPERATING PROCEDURES

OFP C172 FR

Flight Planning Application User Manual

DOCUMENT	OFP-SOP – Application User Manual
REVISION	Rev. 1 – DRAFT for review
DATE	01 OCT 2026
AIRCRAFT	Reims Cessna “Reims Rocket” – I-CCAF (FR172J), I-CCAB (FR172H)
OPERATIONS	VFR flight planning
OUTPUTS	Operational Flight Plan (OFP, A4) · NAV-FLIGHTPLAN (A5)

This manual describes how to operate the application. It does not replace the Aircraft Flight Manual, the Operations Manual or the flight school SOPs. The documents produced are an aid to planning; responsibility for the flight remains with the Pilot in Command.

Contents

	Conventions	3
	Abbreviations	3
SECTION 1	General	5
	1.1 Purpose	5
	1.2 Scope and applicability	5
	1.3 System overview	5
	1.4 Data sources	5
	1.5 Responsibilities	5
	1.6 Data storage	6
SECTION 2	Data Ingestion – Import Page	7
	2.1 Pre-requisites	7
	2.2 Procedure	7
SECTION 3	Pre-flight Data Entry – Wizard	9
	3.0 Wizard navigation	9
	3.1 Step 1 – Flight & Aircraft	9
	3.2 Step 2 – Route & Fuel	11
	3.3 Step 3 – Mass & Balance	13
	3.4 Step 4 – Performance	15
	3.5 Step 5 – Navlog	17
	3.6 Step 6 – Destination Charts	18
	3.7 Step 7 – Briefing	19
	3.8 Step 8 – NOTAM & Weather	20
SECTION 4	System Outputs & Dispatch	22
	4.1 Menu	22
	4.2 Operational Flight Plan (A4)	22
	4.3 NAV-FLIGHTPLAN (A5)	26
	4.4 Dashboard – Saved flights	28
SECTION 5	Operational Notes – Safety & Performance Limits	30
	5.1 Aircraft applicability	30
	5.2 Performance limits	30
	5.3 Fuel planning as implemented	30
	5.4 Navigation data	30
	5.5 Daylight check	31
	5.6 Documents and storage	31
APPENDIX A	Pre-Departure Cross-Check	32

Conventions

PROCEDURE VERBS

LOAD	Upload a file (drag and drop or tap the drop zone).
INPUT	Type a value.
SELECT	Choose from a list, tab, tick box or menu.
VERIFY	Check that a value filled in by the application is present and plausible.
CROSS-CHECK	Compare a value with an independent source (AIP, AFM, briefing, ForeFlight).
CORRECT	Overwrite a value filled in by the application.
REVIEW	Read every message shown and act on each one.

FIELD CLASSIFICATION

☒ M MANDATORY	Pilot input required. Without it the related output is missing or wrong.
☒ A AUTO	Imported or computed by the application. Editable unless shown greyed out. Must be VERIFIED.
☐ O OPTIONAL	Not required for a valid output.

The application does not block progress on empty fields. The classification above is the operating standard of this manual, not a software lock.

FIGURES AND CALLOUTS

Figures are taken from the application in the light theme, with a real flight (LILN – T&G LILE – LILN, alternate LIMF). Lettered markers **A** **B** ... identify the elements described in the legend under each figure and in the procedure that cites it. The figure button ↗ enlarges a figure on screen.

NOTES, CAUTIONS AND WARNINGS

WARNING	Could affect the safety of the flight if ignored.
CAUTION	Could produce a wrong or incomplete document.
NOTE	Additional information.

ABBREVIATIONS

AFM	Aircraft Flight Manual	AIP	Aeronautical Information Publication
ALTN	Alternate aerodrome	ARR / DEST	Arrival / destination aerodrome
ATO	Actual time over	BEW	Basic empty weight
CG	Centre of gravity	C/S	Callsign
DA / PA	Density / pressure altitude	DEP	Departure aerodrome
ECT	End of evening civil twilight	EET	Estimated elapsed time (leg)
ETO	Estimated time over	FF	Fuel flow
FOB	Fuel on board	GS	Ground speed
LDA	Landing distance available	MAP	Manifold pressure
MLAW / MTOW	Maximum landing / take-off weight	OFP	Operational flight plan

PIC	Pilot in command	POB	Persons on board
POH	Pilot's Operating Handbook	SR / SS	Sunrise / sunset
T&G	Touch and go	TAS	True airspeed
TEM	Threat and error management	TOC	Top of climb
TODA / TORA	Take-off distance / run available	TOW	Take-off weight
UTC	Coordinated universal time	ZFW	Zero fuel weight

SECTION 1

General

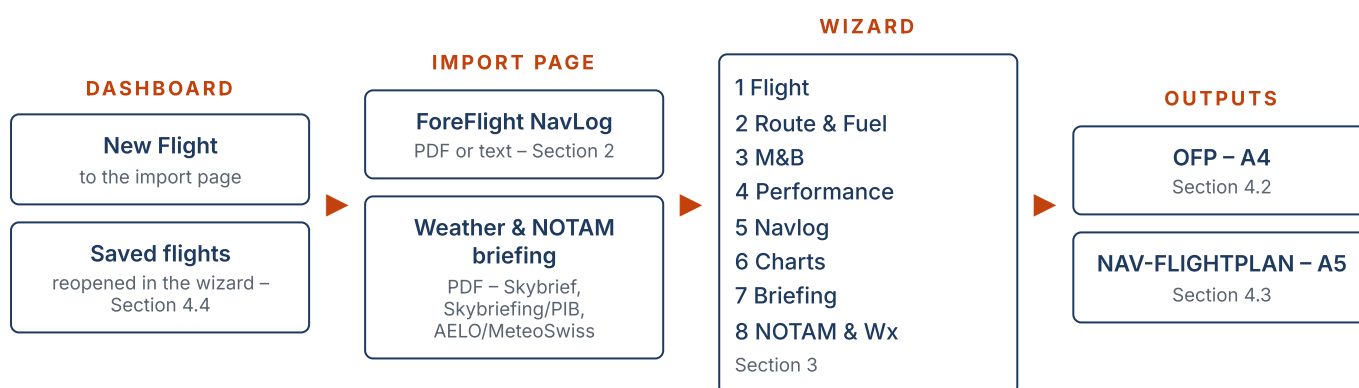
1.1 Purpose

The application converts the route prepared in ForeFlight and the weather briefing into the two documents carried in flight: the **Operational Flight Plan (OFP)** and the **NAV-FLIGHTPLAN**. All performance, fuel and mass figures are computed from the AFM of the selected aircraft; every intermediate result is shown so that it can be checked.

1.2 Scope and applicability

- Aircraft: Reims Cessna "Reims Rocket", Continental IO-360-D: **I-CCAF** (FR172J, AFM Edition 3) and **I-CCAB** (FR172H, AFM Edition 2). The two AFMs differ only in VR, VY, stall speeds and VA: the application uses those of the aircraft selected in Step 1.
- Operations: VFR flight planning. Final reserve is selectable for VFR (30 min) or IFR (45 min).
- Units: fuel in **litres**, fuel flow in **L/h**, speeds in **MPH** (as in the AFM), distances in **NM**, times in **UTC**.

1.3 System overview



1.4 Data sources

DATA	SOURCE
Cruise power, TAS, fuel flow	AFM fig. 5-1, interpolated between 2500 / 5000 / 7500 / 10000 / 15000 ft
Climb time, fuel (incl. start-up and take-off)	AFM fig. 5-6, 1157 kg; climb distance derived from the same data
Take-off / landing distances	AFM fig. 5-4 / 5-5
Basic empty weight, moments, arms	Flying club W. & B. Loading Form (Rev. 17)
Aerodrome position, elevation, runways, frequencies	OurAirports database (physical runway length only)
Route, checkpoints, tracks, distances	ForeFlight NavLog
METAR, SPECI, TAF, NOTAM, SIGMET/AIRMET, charts	Weather briefing PDF

1.5 Responsibilities

The PIC is responsible for the correctness of every input, for every cross-check required by this manual, and for the decision to fly. The application does not navigate, does not decide and does not replace the AFM or the AIP.

1.6 Data storage

Access is restricted: accounts are created by the administrator (Section 2.1). Flights are stored only when the pilot uses **Save to Standby** (Section 4.4). They are saved to the pilot's account, are available on any device after signing in and are not visible to other users. A copy is kept on the device in use, so work can continue without a connection: changes are sent when the connection is restored.

SECTION 2

Data Ingestion – Import Page

2.1 Pre-requisites

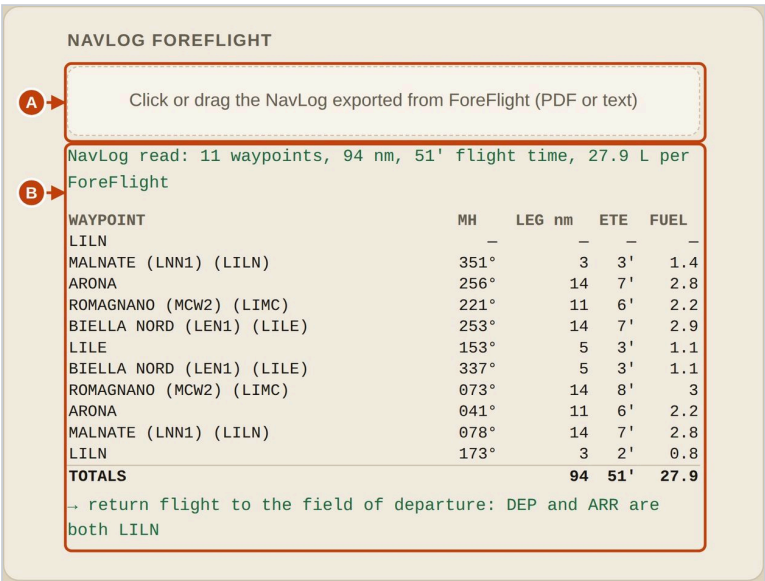
- 1. **INPUT** the e-mail and password provided by the administrator to sign in. At the first sign-in, replace the temporary password with your own (at least 10 characters). The **Dashboard** opens (Section 4.4).
- 2. **PREPARE** the route in ForeFlight and export the **NavLog** (PDF or text).
- 3. **DOWNLOAD** the **weather and NOTAM briefing** PDF covering departure, destination, alternate and route.
- 4. **SELECT New Flight** on the Dashboard: the Import page opens. ← **Dashboard** at the top of the Import page goes back without importing.

2.2 Procedure

- 1. **LOAD** the ForeFlight NavLog in the NavLog drop zone **A**. Load it first: its ICAO codes are used to sort the briefing.
- 2. **VERIFY** the NavLog summary **B**: number of waypoints, total distance and the waypoint table. For a return flight the application states that DEP and ARR are the same aerodrome.

NOTE
Only airports, route and distance are taken from the NavLog. ForeFlight speeds and fuel figures are not used: the AFM figures take precedence.

Figure 2.1 – NavLog Ingestion

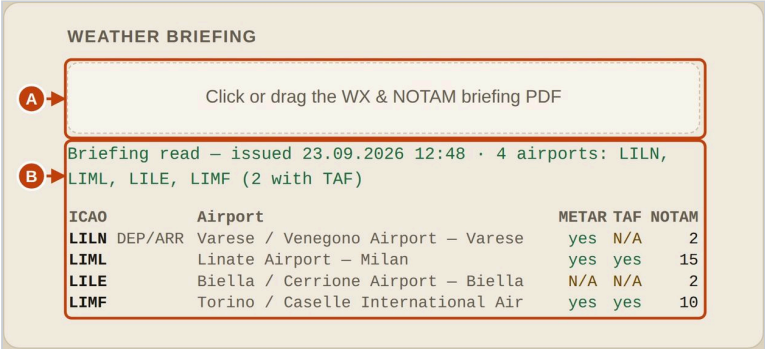


NavLog read: 11 waypoints, 94 NM, return flight to the departure aerodrome.

- A** NavLog drop zone LOAD the ForeFlight NavLog
- B** NavLog summary and waypoint table VERIFY waypoints and distance

- 3. **LOAD** the briefing PDF in the briefing drop zone **A**.
- 4. **VERIFY** the briefing summary **B**: issue date and time (UTC) and, for each aerodrome, the availability of METAR, TAF and the number of NOTAMS. N/A means not contained in the briefing.

Figure 2.2 – Weather Briefing Ingestion



Four aerodromes found in the briefing, two of them with TAF.

- A** Briefing drop zone LOAD the briefing PDF
- B** Briefing summary and aerodrome table VERIFY issue time and coverage

- 5. **CONFIRM** both STATUS items are green **A**
- 6. **SELECT** Continue **B** to open the wizard.

CAUTION
Skip and enter manually **C** opens the wizard without imported data. Route, distances and all weather and NOTAM fields must then be entered by hand.

Figure 2.3 – Import Status



Both files read: the wizard can be opened.

A	Status	CONFIRM both items green
B	Continue	SELECT to proceed
C	Skip and enter manually	Manual entry only

SECTION 3

Pre-flight Data Entry – Wizard

3.0 Wizard navigation

1. **SELECT** a step pill **A** to go to any step, or use **Back** / **Next** at the bottom of the page.
2. **SELECT** the theme button **B** to switch between light and dark display. It does not affect the documents.
3. **SELECT** the menu **C** for documents and saved flights (Section 4.1).

CAUTION
A tick on a step pill only means that the step is behind the current one. It does **not** mean that the step is complete or correct.

Figure 3.1 – Wizard Header



Step pills, theme button and menu.

A Step pills 1-8	Navigate between steps
B Theme button	Light / dark display
C Menu	Documents and flights

3.1 Step 1 – Flight & Aircraft

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
DEP / ARR	A	NavLog	VERIFY
Aerodrome data (name, position, elevation, runways)	A	OurAirports	VERIFY
ALTN	M	Pilot	INPUT
Touch & Go + ICAO	O	Proposed from route	SELECT if landing en route
Aircraft (registration)	M	I-CCAF default; Other = typed by the pilot	SELECT
Flight Mission	O	Pilot	INPUT
Route	A	NavLog	VERIFY
OFF, TO (UTC)	M	Pilot	SELECT
LDG, BLOCK, FLT	A	Computed	VERIFY
ON	A	Proposed, editable	VERIFY / CORRECT

- 1. **VERIFY** DEP and ARR **A** as imported from the NavLog.
- 2. **VERIFY** the aerodrome data line next to each code **B**: name, coordinates, elevation and runways. The line appears automatically when the four-letter code is complete. No line = code not found: CROSS-CHECK the ICAO code.
- 3. **INPUT** the alternate ICAO code **C**. Its weather, NOTAMs, distance and runway data are filled in automatically.
- 4. **SELECT** Touch & Go **D** only if a landing en route is planned. The application proposes the aerodromes found along the route **E**.
- 5. **SELECT** the aircraft **F**. Mass & Balance (Step 3) follows this choice. Select **Other** only for an aircraft not in the list or after a re-weighing: type its registration in the box that appears.
- 6. **INPUT** the flight mission **G** (optional).

- 6. **SELECT** the off-blocks time OFF **A** and the take-off time TO **B**, in UTC, 5-minute steps.
- 7. **VERIFY** LDG **C** = TO + Flight Time computed in Step 2.
- 8. **VERIFY** ON **D** = LDG + the departure taxi time. **CORRECT** it if required: once set by hand it no longer updates; set it back to --:-- to restore automatic mode.
- 9. **VERIFY** BLOCK = ON – OFF and FLT = LDG – TO **E**.
- 10. **REVIEW** the DAYLIGHT check **F**. It requires the flight date (Step 8).

WARNING
Times are **UTC**. Times entered in local time make the daylight check wrong by the UTC offset.

Figure 3.2 – Aerodromes and Mission

Aerodrome codes with the data looked up automatically.

A	DEP / ARR	VERIFY
B	Aerodrome data line	VERIFY elevation and runways
C	ALTN	INPUT
D	Touch & Go	SELECT if applicable
E	Aerodromes along the route	Information
F	Aircraft	SELECT
G	Flight Mission	INPUT (optional)

Figure 3.3 – Estimated Times and Daylight Check

Take-off, landing at destination and landing at the alternate, each checked against its own aerodrome.

A	OFF	SELECT (UTC)
B	TO	SELECT (UTC)
C	LDG	VERIFY
D	ON	VERIFY / CORRECT
E	BLOCK / FLT	VERIFY
F	DAYLIGHT check	REVIEW

DAYLIGHT CHECK – STATUS

INDICATION	MEANING	ACTION
SS -h:mm	Daylight; margin to sunset	VERIFY margin
AFTER SUNSET (+h:mm) / BEFORE SUNRISE	Within civil twilight (amber)	REVIEW plan
NIGHT	After end / before start of civil twilight (red)	REVIEW plan
not computed	Date, TO time, Trip time or coordinates missing	INPUT missing data

3.2 Step 2 – Route & Fuel

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Cruise Altitude	M	Pilot	INPUT
RPM, MAP	M	AFM fig. 5-1 combinations only	SELECT
Trip distance	A	NavLog	VERIFY
Cruise GS	A	TAS of the selected setting (still air)	CORRECT for wind
Trip fuel flow	A	AFM fig. 5-1	VERIFY
Alternate distance	A	Great circle ARR-ALTN	VERIFY / CORRECT
Alternate altitude	M	Pilot	INPUT
Alternate GS, FF, Flight Time	A	22 inHg / 2400 RPM, locked	VERIFY
Flight Rules	M	VFR default	SELECT
Taxi time and fuel	A	Step 1 times, 7.5 L/h	VERIFY
Fuel On Board	O	Pilot (actual uplift)	INPUT
Extra (L)	O	Pilot, if FOB empty	INPUT

- 1. INPUT** the cruise altitude **A**. RPM and MAP stay greyed out until it is entered.
- 2. SELECT** RPM and MAP **B**. Only the combinations published by the AFM at that altitude are offered.
- 3. VERIFY** the power line **C**: % power, TAS and fuel flow.
- 4. VERIFY** the climb line **D**: from DEP elevation to cruise altitude, time, distance and fuel (start-up and take-off included).

Figure 3.4 – Cruise Power Setting

CRUISE POWER SETTING (POH)

Cruise Altitude (ft)	RPM	MAP (inHg)
A 4500	2400	22" — 62% B
C → 22" / 2400 RPM at 4500 ft = 62% power · TAS 132 MPH (115 kt) · 36.3 L/h (interpolated between 2500 and 5000 ft)		
D → climb from 1100 ft to 4500 ft: 5 min, 5.8 NM and 9.50 L at full power, 1157 kg, incl. start-up and takeoff (AFM fig. 5-6)		

Selecting the setting fills in Trip GS and fuel flow.

A Cruise Altitude	INPUT
B RPM / MAP	SELECT
C Power line	VERIFY
D Climb line	VERIFY

5.

VERIFY

the Trip distance

A

from the NavLog.
6.

CORRECT

the cruise GS

B

for the forecast wind. The proposed value is the TAS of the setting.
7.

VERIFY

the fuel flow

C

.
8.

VERIFY

the Trip line

D

: climb + cruise = Trip time and fuel, and the TOC position.

WARNING

An uncorrected GS assumes still air. With a headwind, Trip time and fuel are underestimated.

CAUTION

If the climb is longer than the trip, the line reads **Cruise altitude not reached**. REVIEW altitude or distance.

9.

VERIFY

the alternate distance

A

(great circle ARR–ALTN). CORRECT it for the planned routing.
10.

INPUT

the alternate altitude

B

.
11.

VERIFY

GS and fuel flow

C

, fixed at 22 inHg / 2400 RPM, and the computed Flight Time

D

.
12.

VERIFY

the alternate line

E

: setting, time and fuel.

Figure 3.5 – Trip

TRIP

Distance (nm, from NavLog)

Cruise GS (MPH)

Fuel flow (L/h)

A

94

132

36.3

D

→ climb to 4500 ft: 5.8 NM, 5 min, 9.50 L + cruise 88.2 NM, 46 min at 36.30 L/h = Trip 51 min · 37.41 L · TOC 3 NM after MALNATE

Trip = climb (AFM fig. 5-6) + cruise from the top of climb.

A	Distance	VERIFY
B	Cruise GS	CORRECT for wind
C	Fuel flow	VERIFY
D	Trip line and TOC	VERIFY

Figure 3.6 – Alternate

ALTERNATE

Distance (nm)

GS (MPH)

A

61

132

Fuel flow (L/h)

C

36.3

Altitude (ft)

Flight Time (min, calculated)

B

4500

32

D

E

→ 22" / 2400 RPM = 62% · TAS 132 MPH · 36.3 L/h · time 32 min · fuel 19.36 L

The alternate always uses 22 inHg / 2400 RPM.

A	Distance	VERIFY / CORRECT
B	Altitude	INPUT
C	GS / fuel flow (locked)	VERIFY
D	Flight Time (calculated)	VERIFY
E	Alternate line	VERIFY

13. **SELECT** the flight rules **A**: VFR = 30 min final reserve, IFR = 45 min.
14. **VERIFY** the reserves line **B**: Contingency = max(5% of Trip, 15 min); Final Reserve at cruise fuel flow.
15. **VERIFY** the Plan Block against the 174 L usable **C**.
16. **VERIFY** the taxi time and fuel **D**, derived from OFF-TO and LDG-ON at 7.5 L/h.

CAUTION

ABOVE the 174 L usable on the Plan Block line: the plan exceeds tank capacity.

17. **INPUT** the fuel on board after uplift **A** (optional). It is used only to compute the extra fuel and is not printed.
18. **INPUT** the extra fuel **B** by hand if FOB is left empty.
19. **VERIFY** the line **C**: minimum required, extra litres and minutes.

WARNING

FOB ... INSUFFICIENT in red: the fuel on board is below the minimum required.

Figure 3.7 – Reserves & Margins

RESERVES & MARGINS

Flight Rules

A VFR

B → Contingency = max(5% trip, 15') = 15 min · 9.07 L · Final Reserve = 30 min · 18.15 L

C → Plan Block 120 L = 86.40 kg (31.70 USG) · tank remaining 54 L

Taxi (min, calculated) Taxi (L, calculated)

D 29 3.63

→ 15 min off-blocks-takeoff + 14 min landing-on-blocks = taxi 29 min · 3.63 L at 7.50 L/h

Reserves, tank capacity and taxi are computed, not entered.

A Flight Rules	SELECT
B Contingency / Final Reserve	VERIFY
C Plan Block vs usable fuel	VERIFY
D Taxi	VERIFY

Figure 3.8 – Fuel On Board & Extra

FUEL ON BOARD & EXTRA

Fuel On Board — uplifted (L)

A 120

Extra (L) FT Extra (min)

B 32.38 54

C → minimum required 87.62 L · extra 32.38 L = 54 min at the cruise fuel flow

With FOB entered, the extra is FOB minus the minimum required.

A Fuel On Board	INPUT (optional)
B Extra (L)	INPUT if FOB empty
C Minimum / extra line	VERIFY

3.3 Step 3 – Mass & Balance

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Aircraft	A	Selected in Step 1	VERIFY
Basic Empty Weight, Moment	A	Club W&B form; editable only with Other	VERIFY
Pilot + front pax	M	Pilot	INPUT
Rear pax	O	Pilot	INPUT if carried
Baggage (max 91 kg)	O	Pilot	INPUT if carried
EPOB / POB	M	Pilot	INPUT
Fuel masses	A	Step 2 (1 L = 0.72 kg)	VERIFY
MZFW, MTOW, MLAW	A	AFM limits	VERIFY
Remarks (M&B)	O	Pilot	INPUT

1. **VERIFY** the aircraft **A**, selected in Step 1. Empty weight and moment **B** are filled in and locked. With **Other** (Step 1) they are entered by hand, for example after a re-weighing.

2. **INPUT** pilot and front passenger mass **A**.
3. **INPUT** rear passengers **B** and baggage **C** if carried. Baggage limit 91 kg.
4. **INPUT** persons on board **D**.
5. **VERIFY** the ZFW line **E**.

6. **VERIFY** the mass check **A**: masses within limits.
7. **VERIFY** the CG check **B**: CG at take-off within the envelope and the category available (UTILITY / NORMAL).
8. **VERIFY** the CG envelope chart: take-off, destination landing and alternate landing points inside the envelope.

WARNING

Any mass or CG message other than **within limits** must be resolved before flight.

Figure 3.9 – Aircraft and Empty Weight

AIRCRAFT

Aircraft

A

I-CCAF – CESSNA C172FR (selected in Step 1)

Basic Empty Weight (kg)

Moment BEW (kg·m)

B

734

679

→ empty arm = moment ÷ weight = 0.9251 m

Empty weight and moment from the flying club form.

A	Aircraft (from Step 1)	VERIFY
B	BEW / moment (locked)	VERIFY

Figure 3.10 – Loading

LOADING

Pilot + front pax (kg)

Rear pax (kg)

A

160

kg

B

Baggage (kg, max 91)

EPOB / POB

C

8

2

D

E

→ ZFW = 902 kg · arm 0.941 m · moment 848.68 kg·m

Fuel is not entered here: it comes from Step 2.

A	Pilot + front pax	INPUT
B	Rear pax	INPUT if carried
C	Baggage	INPUT if carried
D	EPOB / POB	INPUT
E	ZFW line	VERIFY

Figure 3.11 – Mass & Balance Result

RESULT

Item	kg	arm m	kg·m
BASIC EMPTY WEIGHT	734	0.925	679
PILOT AND FRONT PASSENGER	160	0.940	150.40
REAR PASSENGERS	0	–	0
BAGGAGE	8	2.410	19.28
ZERO FUEL WEIGHT	902	0.941	848.68
TAXI FUEL	2.61	1.230	3.21
TRIP FUEL	26.94	1.230	33.13
ALTERNATE FUEL	13.94	1.230	17.15
HOLDING / RESERVE	13.07	1.230	16.07
FUEL ON BOARD	86.40	1.230	106.27
RAMP WEIGHT	988.40	0.966	954.95
TAKE OFF WEIGHT	985.79	0.965	951.74
DEST. LDG WEIGHT	958.85	0.958	918.61
ALTN. LDG WEIGHT (no res.)	931.84	0.950	885.39

A

→ masses within limits

B

→ CG at takeoff 0.965 m (limits 0.946–1.2 m) · available category: UTILITY and NORMAL

MZFW

MTOW

MLAW

1157

1157

1157

Loading sheet with the arms of the club form and the AFM limits.

A	Mass check	VERIFY
B	CG and category	VERIFY

3.4 Step 4 – Performance

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Display Unit	☐	MPH default	SELECT (display only)
Runway Surface	☒	Asphalt, dry default	SELECT
Runway (each aerodrome)	☒	None pre-selected	SELECT
Elevation, RWY heading	☐	OurAirports	VERIFY
TORA / TODA / LDA	☐	OurAirports physical length	CROSS-CHECK with AIP
QNH, Temp, Wind	☐	Own METAR (or nearest station)	VERIFY
Runway (OFP)	☐	Runway selected	VERIFY

1. **SELECT** the display unit **A** (MPH or knots). Calculations do not change.
2. **SELECT** the runway surface **B**: asphalt dry, grass dry, or wet / contaminated.

CAUTION
Wet / contaminated: the AFM publishes no data and no increment is applied. A red warning is shown.

Figure 3.12 – Speeds and Surface

SPEEDS

Display Unit

A

MPH

▼

Runway Surface

B

Asphalt, dry

▼

A

Display Unit

SELECT

B

Runway Surface

SELECT

Each phase has its own tab **A**: **DEP** (take-off), **ARR** (landing), **ALT** (alternate landing) and **ENR** (touch and go, shown only when selected in Step 1). A dot on a tab indicates a warning for that phase.

3. **SELECT** each tab **A** in turn and carry out items 4 to 8.

4. **SELECT** the runway **B**. No runway is pre-selected: select it from the wind of the day.

5. **VERIFY** the elevation **C** and the runway heading.

6. **CROSS-CHECK** TORA and TODA (LDA for landing) **D** with the AIP. They are pre-filled with the physical runway length.

7. **VERIFY** QNH, temperature and wind **E**, and their source **F**.

8. **VERIFY** the result card **G**: pressure and density altitude, head/crosswind against the limit, distances required and margin.

Uncontrolled copy when printed

Page 15 of 32

Figure 3.13 – Take-off and Landing Data – DEP Tab

DEPARRALTENR

DEPARTURE — TAKEOFF

Airport

LILN

Runway

17 — 1420 m A

Runway (OFP)

17

Elevation (ft)

1100

RWY Heading (°M)

170

TORA (m)

1420

TODA (m)

1420

QNH (hPa)

1023

Temp (°C)

22

Wind From (°M)

190

Wind (kt)

5

QNH, temperature and wind from the LILN METAR.

DEP TAKEOFF — LILN 17 · 968 KG

Flap 10° · VR 62 MPH · VY 82 MPH

PRESSURE ALTITUDE

837 ft

DENSITY ALTITUDE

1865 ft

ISA DEVIATION

+8.7 °C

HEADWIND

4.7 kt

CROSSWIND

1.7 kt from the right

T/O limit 20 ktQNH 1023 · 22°C

Takeoff run

TORR · TO LIFT-OFF

145 m476 ft

Within limits — 1275 m margin, 10% of 1420 m

Takeoff distance

TODR · TO CLEAR 15 M

262 m860 ft

Within limits — 1158 m margin, 18% of 1420 m

Inputs on the left, result card on the right. Wind, QNH and temperature come from the LILN METAR.

A Phase tabs DEP / ARR / ALT / ENR	SELECT each	B Runway	SELECT
C Elevation	VERIFY	D TORA / TODA	CROSS-CHECK with AIP
E QNH / Temp / Wind	VERIFY	F Weather source	VERIFY
G Result card	VERIFY margins		

9. **REVIEW** every message in **Checks & Warnings** **A**. Red = the result does not cover the condition (e.g. tailwind, wet runway, standard weather used).

- NOTE**

An aerodrome without METAR uses standard values (1013 hPa, 15 °C, calm) and is marked in red. If a station with METAR exists within 30 NM, a button offers its values.
- NOTE**

Touch and go: landing and take-off are both computed with the take-off mass (conservative).

Figure 3.14 – Checks & Warnings

CHECKS & WARNINGS

TAILWIND

T&G landing — Landing: the AFM publishes no tailwind data. The distances are for zero wind.

TAILWIND

T&G takeoff — Takeoff: the AFM publishes no tailwind data. The distances shown are for zero wind and are shorter than the actual ones.

WEATHER

LILE: QNH, temperature and wind taken from the LIMF METAR, 26 NM away — this airport has no station.

LANDING MASS

The landing table is published at 1157 kg only and is used whatever the mass: at lower masses it is conservative.

Example: tailwind on the T&G runway and weather taken from a nearby station.

A Checks & Warnings	REVIEW every item
----------------------------	-------------------

3.5 Step 5 – Navlog

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
GS / fuel flow navplan	☐	Step 2 values if empty	INPUT only to override
Freq.	☒	OurAirports (number only)	VERIFY
C/S	☐	Pilot	INPUT
Checkpoint, MT, Dist.	☒	NavLog	VERIFY
Alt.	☒	Cruise altitude	VERIFY / CORRECT per waypoint
EET, ETO, Fuel	☒	Computed	VERIFY
Remarks	☐	Pilot	INPUT

1. **VERIFY** GS and fuel flow **A** **B**. Leave empty to use the Step 2 values.

Figure 3.15 – Navlog Speed & Fuel Flow

SPEED & FUEL FLOW

GS navplan (MPH)

A

from power setting

Fuel flow navplan (L/h)

from power setting

B

A GS navplan

Leave empty or INPUT

B Fuel flow navplan

Leave empty or INPUT

2. **SELECT + Row** **A** to add a checkpoint, x to delete one.
3. **SELECT Fill Frequencies** **B** to repeat the frequency look-up. **VERIFY** frequencies **C**.
4. **INPUT** callsigns **D** (not in any database).
5. **VERIFY** checkpoints, magnetic tracks and distances **E**.
6. **VERIFY** the altitude of each waypoint **F**; **CORRECT** where different from cruise.
7. **VERIFY** EET per leg, ETO (minutes after TO) and fuel **G**. A leg from or to an aerodrome reads e.g. 7/+5 : 7 min en route + 5 aerodrome minutes (Section 5.3), already included in ETO, Trip and fuel.
8. **INPUT** remarks **H** (optional).
9. **CROSS-CHECK** the totals line **I** with the Trip of Step 2: time and fuel must be the same.

Figure 3.16 – Navlog Legs

LEGS

A

+ Row

B

Fill Frequencies

C

FREQ.

D

C/S

E

CHECKPOINT

MT

F

DIST.

ALT.

G

EET ETO FUEL

H

REMARKS

128.55	VENEGO	LILN			4500	0' 45' 0 L		x
		MALNATE	351	3	4500	2' 47' 4.92 L		x
		ARONA	256	14	4500	8' 55' 8.13 L		x
		ROMAGNANO	221	11	4500	6' 01' 3.48 L		x
		BIELLA NORD	253	14	4500	7' 08' 4.43 L		x
123.05	BIELLA	LILE	153	5	4500	3' 11' 1.58 L	T&G	x
		BIELLA NORD	337	5	4500	3' 14' 1.58 L		x
		ROMAGNANO	073	14	4500	7' 21' 4.43 L		x
		ARONA	041	11	4500	6' 27' 3.48 L		x
		MALNATE	078	14	4500	7' 34' 4.43 L		x
128.55		LILN	173	3	4500	2' 36' 0.95 L		x

I

→ 11 legs · 94 nm · 51' · 37.41 L at GS 132 MPH and 36.30 L/h

Times and fuel are recomputed from the selected power setting, not copied from ForeFlight; the climb is split over the legs by distance.

A + Row	Add checkpoint	B Fill Frequencies	Repeat look-up
C Freq.	VERIFY	D C/S	INPUT
E Checkpoint / MT / Dist.	VERIFY	F Alt.	VERIFY / CORRECT
G EET / ETO / Fuel	VERIFY	H Remarks	INPUT (optional)
I Totals line	CROSS-CHECK with Trip		

NOTE

A leg longer than 10 minutes shows an advisory message (turning points ideally every 6–8 minutes). It does not block the wizard and does not change the navlog.

3.6 Step 6 – Destination Charts

1. **LOAD** up to 2 images or PDFs (aerodrome or approach charts) (optional).
2. **INPUT** the operational notes as required: Track, Runway, Altitudes, Clearance, Timing, On ground, Radios, Specials (optional).

Charts and notes are printed on the DEST CHARTS page of the OFP. Graphic pages of the briefing (e.g. significant weather, wind charts) are added to the OFP automatically, one per page.

3.7 Step 7 – Briefing

A briefing by phase of flight, written before departure, that gives a structure to everything that has to be done in flight. All fields are optional free text and nothing is checked by the application; they are printed on the BRIEFING page of the OFP (Section 4.2).

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Procedure (DEP, ENR, T&G, DEST, ALTN)	☐	Pilot	INPUT
Possible threats	☐	Pilot	INPUT
Mitigation	☐	Pilot	INPUT
Aircraft status	☐	Pilot	INPUT if equipment is inoperative
Remarks	☐	Pilot	INPUT if applicable

1.

INPUT

for the phase the procedure **A**:

what will be flown, e.g. runway, routing, reporting points, altitudes, frequencies.
2.

INPUT

the possible threats **B** of that

phase, e.g. weather, terrain, traffic, airspace, aircraft, pilot.
3.

INPUT

the mitigation **C**: what will be done

to keep each threat under control.
4.

INPUT

the same three fields for Enroute, Touch & Go, Destination and Alternate. The Touch & Go block is shown only when Touch & Go is selected in Step 1; each heading shows the aerodrome code.

Figure 3.17 – Briefing – Departure

DEPARTURE LILN

Dep procedure

A

RWY 35: climb on runway heading to 1500 ft, exit via the mandatory VFR reporting point, then climb to 4500 ft. Squawk 7000.

Poss. threats

B

Traffic converging on the reporting point; obstacles on the climb-out.

Mitigation

C

Listen out on the frequency before line-up, landing light ON; climb gradient checked against the obstacles.

Example. Every phase has the same three fields.

A	Procedure	INPUT
B	Possible threats	INPUT
C	Mitigation	INPUT

1.

INPUT Aircraft status **A** only when

instruments or equipment are temporarily inoperative, stating which aids will therefore not be available in flight.
2.

INPUT Remarks **B** only for important

information on the conduct of the flight that is not related to the routing: exam flight, solo flight, dual command, SPIC flight.

NOTE

Keep the text short and in the order it will be used: the BRIEFING page is read in flight.

Figure 3.18 – Aircraft Status and Remarks

AIRCRAFT STATUS

A

ADF INOP (placarded): NDB not usable.

Instruments or equipment temporarily inoperative, and therefore aids that will not be available in flight (e.g. ADF INOP: NDB not usable).

REMARKS

B

Solo flight.

Important information on the conduct of the flight that has nothing to do with the routing: exam flight, solo flight, dual command, SPIC flight.

Example.

A	Aircraft status	INPUT if applicable
B	Remarks	INPUT if applicable

3.8 Step 8 – NOTAM & Weather

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Briefing PDF	A	Loaded at import; can be reloaded	LOAD only to update
Date (DD.MM.YYYY)	M	Briefing	VERIFY / INPUT
Time (UTC), Validity	A	Briefing	VERIFY
METAR, SPECI, TAF, NOTAM per aerodrome	A	Briefing	VERIFY
Enroute NOTAMs, SIGMET / AIRMET	A	Briefing	VERIFY
PIC name	M	Pilot	INPUT

1.

LOAD

a newer briefing

A

only to update the weather.
2.

VERIFY

the status line

B

: aerodromes found and those not on the route.
3.

VERIFY

the date

C

. It is printed on the OFP and used by the daylight check. INPUT it if no briefing was loaded.
4.

VERIFY

issue time

D

and validity

E

.

NOTE

If the briefing was issued on the day before the flight, the date shown is the issue date.

Figure 3.19 – Briefing Data

A

Click or drag the WX & NOTAM briefing PDF

B

Briefing read (skybrief) – airports found: LILN, LIML, LILE, LIMF · 3 fields filled in · off route: LIML

C

Date (DD.MM.YYYY)

23.09.2026

Time (HH:MM UTC)

12:48

D

E

Validity / period

extracted from the PDF, if present

A	Briefing drop zone	LOAD to update
B	Status line	VERIFY
C	Date	VERIFY / INPUT
D	Time (UTC)	VERIFY
E	Validity	VERIFY

5. **VERIFY** for each aerodrome (DEP, ARR, T&G, ALTN) METAR **A**, SPECI **B**, TAF **C** and NOTAM **D**: correct aerodrome, current issue.
6. **VERIFY** the Enroute NOTAM and SIGMET / AIRMET boxes. An absence must be stated (e.g. "no active SIGMET"): an empty box is not information.
7. **INPUT** the PIC name. It is printed on every page of the OFP and on the NAV-FLIGHTPLAN.

NOTE

N/A – not available in the briefing is an explicit statement that the report is not contained in the briefing.

Figure 3.20 – Aerodrome Weather & NOTAM

DEPARTURE AIRPORT (LILN)

METAR

A METAR LILN 231220Z AUTO 19005KT //// // 22/10 Q1023=

SPECI

B

TAF

C N/A – not available in the briefing

NOTAM

D C0792/26 27 Jul 12:00Z – 03 Oct 16:00Z TESTO ITALIANO: CAUSA PRESENZA GRU LE DISTANZE DICHIARATE CAMBIANO COME SEGUE: RWY17: TORA 1118M, TODA 1142M, ASDA 1176M, LDA

One block per aerodrome. Text edited by hand is not overwritten by a later import.

A	METAR	VERIFY
B	SPECI	VERIFY
C	TAF	VERIFY
D	NOTAM	VERIFY

SECTION 4

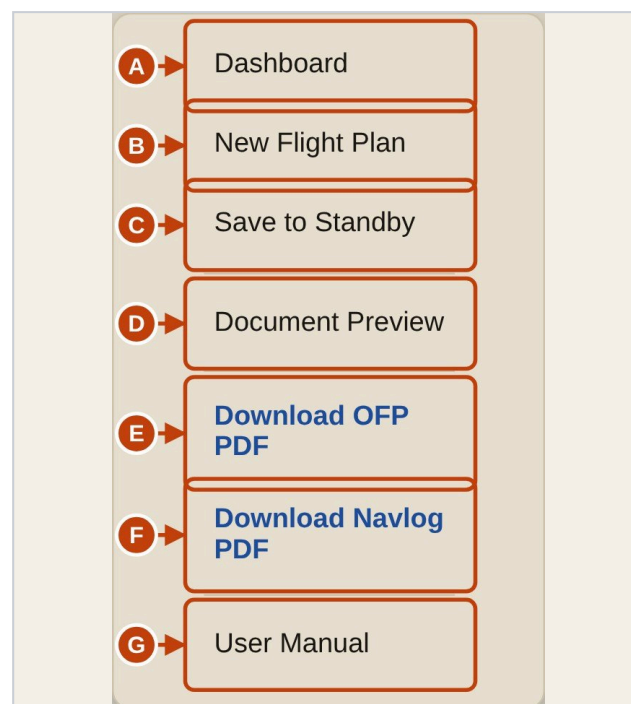
System Outputs & Dispatch

4.1 Menu

ITEM	FUNCTION
A Dashboard	Back to the Dashboard (Section 4.4). The flight plan stays open.
B New Flight Plan	Closes the flight plan and opens the import page. Unsaved data is lost.
C Save to Standby	Saves the flight to your account
D Document Preview	Shows the OFP on screen
E Download OFP PDF	A4 OFP. To print it, use the print function of the PDF viewer.
F Download Navlog PDF	A5 NAV-FLIGHTPLAN (separate document)
G User Manual	This manual, in a new tab

Below these items the menu shows the signed-in user, **Users** (administrator only), **Change password** and **Sign out**. Users and Change password open in a new tab. On the Dashboard the menu shows only New Flight Plan, User Manual and the account items. Below Download Navlog PDF, **Download Club Navlog PDF** produces the flying club form (Section 4.3).

Figure 4.1 – Main Menu



Menu at the top right of the wizard.

4.2 Operational Flight Plan (A4)

OFP PAGE STRUCTURE (NUMBER OF PAGES VARIES WITH NOTAMS AND CHARTS)

PAGE	CONTENT
OFP	Flight data, estimated / actual times, route, cruise altitude, fuel plan, alternate summary, daylight check
MASS AND BALANCE	Loading sheet, limits, CG envelope chart, remarks, PIC signature
PERFORMANCE	Masses, V-speeds, take-off and landing distances and margins, cruise power setting, TOC, AFM cruise table
DEST CHARTS	Charts and operational notes (Step 6)
BRIEFING	Briefing by phase (DEP, ENR, T&G, DEST, ALTN): procedure, threats, mitigation; aircraft status, remarks (Step 7)
WEATHER	METAR / SPECI / TAF by aerodrome, SIGMET / AIRMET
NOTAM	NOTAM by aerodrome and enroute
Charts	Graphic pages of the briefing, if any

Figure 4.2 – OFP – Page 1

LILN-LILN CESSNA C172 FR I-CCAF																														
OFP																														
A → FLT MISSION: VFR navigation training DEP: LILN T&G: LILE ARR: LILN ALTN: LIMF	<table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 30%;"></th> <th style="width: 30%; text-align: center;">ESTIMATED</th> <th style="width: 40%; text-align: center;">ACTUAL</th> </tr> </thead> <tbody> <tr> <td>OFF</td> <td style="text-align: center;">0830</td> <td style="text-align: center;">.....Z</td> </tr> <tr> <td>TO</td> <td style="text-align: center;">0845</td> <td style="text-align: center;">.....Z</td> </tr> <tr> <td>LDG</td> <td style="text-align: center;">0936</td> <td style="text-align: center;">.....Z</td> </tr> <tr> <td>ON</td> <td style="text-align: center;">0950</td> <td style="text-align: center;">.....Z</td> </tr> <tr> <td>BLOCK</td> <td style="text-align: center;">0120</td> <td style="text-align: center;">.....Z</td> </tr> <tr> <td>FLT</td> <td style="text-align: center;">0051</td> <td style="text-align: center;">.....Z</td> </tr> </tbody> </table>				ESTIMATED	ACTUAL	OFF	0830	Z	TO	0845	Z	LDG	0936	Z	ON	0950	Z	BLOCK	0120	Z	FLT	0051	Z						
	ESTIMATED	ACTUAL																												
OFF	0830	Z																												
TO	0845	Z																												
LDG	0936	Z																												
ON	0950	Z																												
BLOCK	0120	Z																												
FLT	0051	Z																												
C → ROUTE: MALNATE ARONA ROMAGNANO BIELLA NORD LILE BIELLA NORD ROMAGNANO ARONA MALNATE CRUISE ALTITUDE: 4500 ft																														
<table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 60%;"></th> <th style="width: 20%; text-align: center;">FUEL(L)</th> <th style="width: 20%; text-align: center;">TIME</th> </tr> </thead> <tbody> <tr> <td>TRIP</td> <td style="text-align: center;">37.41</td> <td style="text-align: center;">51</td> </tr> <tr> <td>CONT</td> <td style="text-align: center;">9.07</td> <td style="text-align: center;">15</td> </tr> <tr> <td>ALTN</td> <td style="text-align: center;">19.36</td> <td style="text-align: center;">32</td> </tr> <tr> <td>FINAL RES</td> <td style="text-align: center;">18.15</td> <td style="text-align: center;">30</td> </tr> <tr> <td colspan="3" style="border-top: 1px dashed black; padding-top: 5px;"> D → MIN BLOCK 83.99 128 </td> </tr> <tr> <td>EXTRA</td> <td style="text-align: center;">32.38</td> <td style="text-align: center;">54</td> </tr> <tr> <td>TAXI</td> <td style="text-align: center;">3.63</td> <td style="text-align: center;">29</td> </tr> <tr> <td colspan="3" style="border-top: 1px dashed black; padding-top: 5px;"> PLAN BLOCK 120 211 </td> </tr> </tbody> </table>					FUEL(L)	TIME	TRIP	37.41	51	CONT	9.07	15	ALTN	19.36	32	FINAL RES	18.15	30	D → MIN BLOCK 83.99 128			EXTRA	32.38	54	TAXI	3.63	29	PLAN BLOCK 120 211		
	FUEL(L)	TIME																												
TRIP	37.41	51																												
CONT	9.07	15																												
ALTN	19.36	32																												
FINAL RES	18.15	30																												
D → MIN BLOCK 83.99 128																														
EXTRA	32.38	54																												
TAXI	3.63	29																												
PLAN BLOCK 120 211																														
E → BLOCK = ACT USD=																														
F → PLAN BLOCK = 120 L = 86.40 kg (31.70 USG) USABLE = 174 L (46 USG)																														
<table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 35%;"></th> <th style="width: 20%; text-align: center;">DIST</th> <th style="width: 20%; text-align: center;">TIME</th> <th style="width: 25%; text-align: center;">ALTITUDE</th> </tr> </thead> <tbody> <tr> <td>G → ALTERNATE LIMF</td> <td style="text-align: center;">61</td> <td style="text-align: center;">32</td> <td style="text-align: center;">4500</td> </tr> </tbody> </table>					DIST	TIME	ALTITUDE	G → ALTERNATE LIMF	61	32	4500																			
	DIST	TIME	ALTITUDE																											
G → ALTERNATE LIMF	61	32	4500																											
H → DAYLIGHT 23.09.2026 (UTC) TO 0845Z LILN SR 0513 SS 1720 ECT 1750 SS -8:35 LDG 0936Z LILN SR 0513 SS 1720 ECT 1750 SS -7:44 ALTN 1008Z LIMF SR 0518 SS 1725 ECT 1754 SS -7:17																														

The ACTUAL column and the BLOCK / ACT USD lines are completed by hand.

A Flight data	DEP / T&G / ARR / ALTN	B Estimated / actual times	Record ACTUAL in flight (UTC)
C Route and cruise altitude	VERIFY	D Fuel plan	CROSS-CHECK with Step 2
E BLOCK / ACT USD	Complete by hand	F Plan block mass, usable fuel	VERIFY
G Alternate summary	VERIFY	H Daylight check	REVIEW

Figure 4.3 – OFP – Mass and Balance

MASS AND BALANCE			
A/C I-CCAF POB 2			
	WEIGHT kg	ARM m	MOMENT kg*m
WEIGHT CALCULATION			
A → BASIC EMPTY WEIGHT	734	0.925	679
PILOT AND FRONT PASSENGER	160	0.940	150.40
REAR PASSENGERS	0	--	0
BAGGAGE	8	2.410	19.28
ZERO FUEL WEIGHT	902	0.941	848.68
FUEL CALCULATION			
B → TAXI FUEL	2.61	1.230	3.21
TRIP FUEL	26.94	1.230	33.13
ALTERNATE FUEL	13.94	1.230	17.15
HOLDING / RESERVE	13.07	1.230	16.07
FUEL ON BOARD	86.40	1.230	106.27
RESULT			
C → RAMP WEIGHT	988.40	0.966	954.95
TAKE OFF WEIGHT	985.79	0.965	951.74
DEST. LDG WEIGHT	958.85	0.958	918.61
ALTN. LDG WEIGHT (no res.)	931.84	0.950	885.39
D → LIMITS: MTOW/MLW 1157 kg - BAGGAGE 91 kg - FUEL 174 L (46 USG) CG T/O 0.965 m - NORMAL envelope 0.946-1.2 m - WITHIN LIMITS CATEGORY AVAILABLE: UTILITY and NORMAL - VA at 986 kg = 110 MPH			

Followed on the same page by the CG envelope chart, remarks and PIC signature.

A Weight calculation	VERIFY	B Fuel calculation	VERIFY
C Result	VERIFY	D Limits and CG	VERIFY within limits

Figure 4.4 – OFP – Performance

PERFORMANCE

TOW 968 kg LAW 941 kg ALTN 914 kg ← **A**

*ADD +5 KTS AT VREF SPEEDS

VNE 161 VNO 127 VFE 87 KIAS XWIND T/O 20 kt LDG 15 kt

VA 120 MPH at 1157 kg -> 109 MPH at 968 kg (TOW)

B →

MPH	DEP	DEST	ALTN
VR - 15 m	62	-	-
VY best rate	82	-	-
V best glide	85	85	85
Vref flap 40	-	73	73

AFM PUBLISHED SPEEDS (MPH)

	850 kg	1000 kg	1157 kg
VR - 15 m	58	63	68
VY at SL	80	82	85
ROC SL ft/min	1420	1120	880
V best glide	85 MPH		
Vref flap 40	73 MPH		
Stall UP/20/40	61 / 55 / 53 MPH		

TAKE-OFF & LANDING DISTANCES

C →

PHASE	AIRPORT/RWY	MASS	QNH / T / WIND	PA / DA	HW / XW	GND ROLL	15 m
T/O DEP	LILN 17	968 kg	1023 / 22C / 190-5kt	837 / 1865 ft	HW4.7 / XW1.7	145 m	262 m
LDG T&G	LILE 16	968 kg	1024 / 22C / 0-3kt	630 / 1609 ft	TW2.8 / XW1	192 m	393 m
T/O T&G	LILE 16	968 kg	1024 / 22C / 0-3kt	630 / 1609 ft	TW2.8 / XW1	168 m	293 m
LDG DEST	LILN 17	941 kg	1023 / 22C / 190-5kt	837 / 1865 ft	HW4.7 / XW1.7	175 m	358 m
LDG ALTN	LIMF 36	914 kg	1024 / 22C / 0-3kt	699 / 1695 ft	HW3 / XW0	181 m	370 m

D →

PHASE	DIST	AVAILABLE	REQUIRED	USED	MARGIN
T/O DEP	TORA	1420 m	145 m	10%	1275 m
T/O DEP	TODA	1420 m	262 m	18%	1158 m
LDG T&G	LDA	1320 m	393 m	30%	927 m
T/O T&G	TORA	1320 m	168 m	13%	1152 m
T/O T&G	TODA	1320 m	293 m	22%	1027 m
LDG DEST	LDA	1420 m	358 m	25%	1062 m
LDG ALTN	LDA	3300 m	370 m	11%	2930 m

CRUISE POWER SETTING - 4500 ft

E →

PHASE	ALTITUDE	MAP	RPM	POWER	TAS	FUEL FLOW	TIME / L
CLIMB	1100-4500 ft	-	-	full	-	-	5 min / 9.50 L
CRUISE	4500 ft	22"	2400	62%	132 MPH	36.3 L/h	-
ALTERNATE	4500 ft	22"	2400	62%	132 MPH	36.3 L/h	32 min / 19.36 L

TOC 3 NM after MALNATE (climb 6 NM) ← **F**

G →

2200 RPM				2300 RPM				2400 RPM				2500 RPM				2600 RPM			
MP	%	MPH	L/h	MP	%	MPH	L/h	MP	%	MPH	L/h	MP	%	MPH	L/h	MP	%	MPH	L/h
23	58	128	33.8	23	62	132	36.3	23	67	136	38.6	23	71	140	41.2	23	74	143	42.9
22	54	123	31.8	22	58	128	33.8	22	62	132	36.3	22	67	136	38.6	22	70	139	40.1
21	50	119	29.6													21	65	135	37.5
20	46	113	27.7																
19	43	107	25.8																
18	39	99	23.9																

A Masses
(TOW / LAW /
ALTN)

VERIFY

B V-speeds by
aerodrome

VERIFY

C Take-off and
landing
distances

VERIFY

D Distances
available /
required /
margin

CROSS-CHECK

E Cruise power
setting

VERIFY

F TOC position

VERIFY

G AFM cruise
table

Reference

4.3 NAV-FLIGHTPLAN (A5)

1. **SELECT** Download Navlog PDF from the menu. It is a separate document and never part of the OFP.
2. **VERIFY** the header **A**: aircraft, pilot, date, GS, QNH and runway.
3. **RECORD** off-blocks and on-blocks times **B** by hand.
4. **VERIFY** the legs **C**.
5. **RECORD** ATO **D** in flight at each checkpoint.
6. **VERIFY** the alternate row **E** and the fuel calculation box **F**.

Figure 4.5 – NAV-FLIGHTPLAN

NAV-FLIGHTPLAN																	
ACFT IDENT: I-CCAF			Date: 23.09.2026		Off Bl.: QNH: 1023												
Pilot: PIC NAME			GS: 115kt		Bl. on:		RWY: 17										
Freq.	C/S	Checkpoint	MT	Dist.	Alt.	EET	ETO	ATO	Remarks								
128.55	VENEGO	LILN			4500	0'	45'										
		MALNATE	351	3	4500	2'	47'										
		ARONA	256	14	4500	8'	55'										
		ROMAGNANO	221	11	4500	6'	01'										
		BIELLA NORD	253	14	4500	7'	08'										
123.05	BIELLA	LILE	153	5	4500	3'	11'		T&G								
		BIELLA NORD	337	5	4500	3'	14'										
		ROMAGNANO	073	14	4500	7'	21'										
		ARONA	041	11	4500	6'	27'										
		MALNATE	078	14	4500	7'	34'										
128.55		LILN	173	3	4500	2'	36'										
		Alternate: LIMF		61	4500	32'											
						Fuel calc. I/h: 36 Time Fuel											
						Trip				00:51				37			
						Alternate				00:32				19			
						Reserve				00:30				18			
						Minimum				02:08				83			
						Extra fuel				00:54				32			
						Block fuel				03:31				120			

A5 document for the cockpit.

A Header	VERIFY	B Off Bl. / Bl. on	Record by hand
C Legs	VERIFY	D ATO	Record in flight
E Alternate row	VERIFY	F Fuel calculation	VERIFY

Download Club Navlog PDF produces the same navigation log on the flying club form (Aeroclub di Varese, Navigation Log): one PDF of two A4 landscape pages, front and back, to be printed double-sided. It does not replace the NAV-FLIGHTPLAN, which remains the navigation log of the application.

1. **SELECT** Download Club Navlog PDF from the menu.
2. **VERIFY** the main table (right side of the front): checkpoints, NAV AIDS (Ident = callsign, Freq), MC, ALT, DIST and FUEL per leg and remaining, ETE and ETO, as in the NAV-FLIGHTPLAN. Up to 13 checkpoints fit; up to 6 more continue in the left table, above the Dest. Airport row.
3. **VERIFY** the alternate (left side of the front): one leg from destination to alternate. The MC is computed from the great circle and the WMM2025 magnetic variation, and is marked as to be verified.
4. **VERIFY** the back: weather by aerodrome from the METAR; weight and balance with the take-off fuel; fuel calculation; take-off data (DEP) and landing data (DEST).
5. **RECORD** block times, RETO and ATO by hand.

NOTE

Differences from the club form as printed by hand: CLIMB fuel is the AFM fig. 5-6 block (start-up and take-off included), so no rate is given; DESC. is not published in the AFM; APP. is the 5 minutes per aerodrome; the contingency, which the form does not have, is included in MINIMUM FOB and stated in the Note. ASDA (required) is the take-off ground roll plus the landing ground roll. Emax is the best glide speed. ASDA declared, VX, VLO and VLE are left blank.

CAUTION

The form is an aid to flight preparation and does not replace the AFM, the SOP or the AIP. Every pre-filled value must be verified by the pilot before use.

Figure 4.6 – Club Navlog – Front

[illegible]

1

Main table on the right, alternate on the left.

Figure 4.7 – Club Navlog – Back

WEATHER BRIEFING						
STATION	TIME	WIND	VISIBILITY	CLOUDS	TEMP.	QNH
LILN	0720Z	170 / 08 KT	CAVOK	CAVOK	16/07	1018
LILE	0720Z	150 / 06 KT	CAVOK	CAVOK	17/06	1018
LIMF	0720Z	190 / 07 KT	9999	FEW040	18/08	1017
		/				
		/				
		/				

WEIGHT AND BALANCE			
	WEIGHT	X	ARM = MOMENT
AIRCRAFT EMPTY WEIGHT	734.0 kg		0.925 m
FRONT PASSENGERS	165.0 kg		0.940 m
REAR PASSENGERS	75.0 kg		1.860 m
FUEL	98.6 kg		1.230 m
BAGGAGE	12.0 kg		2.410 m
ATOW	1084.6 kg		TOTAL MOMENT
			1123.8 kg m

TAKE OFF DATA			
FIELD ELEV.	RWY	RWY COND.	FLAP
1100 ft	17	ASPHALT, DRY	VR
QNH	O.A.T.	WIND	VX
1018 hPa	16°C	170/08 KT	
M.T.O.W.	A.T.O.W.	WIND/C	VY
1157 kg	1085 kg	H 8.0 X 0.0	84
	Va	114	VFE
	Emax	85	VLO
TORA (Declared)	TODA (Declared)	ASDA (Declared)	LDA (Declared)
1420 m	1420 m		1420 m
Ground roll	T/off distance	ASDA	Landing Distance
163 m	289 m	326 m	333 m

FUEL CALCULATION					
DESTINATION			ALTERNATE		
Rate	Time in Minutes	Gall./LT. Required	Rate	Time in Minutes	Gall./LT. Required
CLIMB AFM	5	9.50	CLIMB	-	-
CRUISE 36.30	46	27.85	CRUISE 36.30	32	19.36
DESC.	-	-	DESC.	-	-
APP 36.30	20	12.10	APP 36.30	5	3.03
Total to Destination		49.45	Total to Alternate		22.39
Note: Fuel in litres (1 L = 0.72 kg). CLIMB: AFM fig. 5-6, start-up and take-off included. APP.: 5 min per aerodrome. CONTINGENCY 15 min, 9.07 L: included in MINIMUM FOB. MINIMUM FOB = DEST + CONT + ALTN + FINAL RES + TAXI W&B fuel: take-off fuel 137 L = 98.64 kg.			FINAL RESERVE 30 MIN		
			TAXI/RUN-UP		
			MINIMUM FOB		
			ACTUAL FOB		
			EXTRA FOB		
			18.15		
			3		
			102.06		
			140		
			37.94		

LANDING DATA			
FIELD ELEV.	RWY	RWY COND.	FLAP
1100 ft	17	ASPHALT, DRY	Vapp
QNH	O.A.T.	WIND	VFE
1017 hPa	17°C	180/10 KT	
M.L.W.	A.L.W.	WIND/C	VLO
1157 kg	1049 kg	H 9.8 X 1.7 R	
	Va	112	VLE
	Emax	85	Vref Land
TORA (Declared)	TODA (Declared)	ASDA (Declared)	LDA (Declared)
1420 m	1420 m		1420 m
Ground roll	T/off distance	ASDA	Landing Distance
156 m	257 m	297 m	318 m

1

Weather, weight and balance, fuel calculation, take-off and landing data.

4.4 Dashboard – Saved flights

The Dashboard is the first page after signing in. **New Flight** opens the Import page (Section 2); saved flights are reopened from their cards. While a flight plan is open in the wizard, **Back to the open flight plan** returns to it. Reloading the page also shows the Dashboard first.

1. **SELECT** **Save to Standby** to store the flight, for example while waiting for the weather. The Dashboard opens.
2. **LOAD** the briefing PDF on the standby card **B** to finalise the flight: it moves to **READY TO FLY** **D**.
3. **SELECT** **Edit data** **C** to reopen a standby flight in the wizard.
4. **LOAD** a newer briefing on a completed card **E** to update its weather.
5. **SELECT** **View OFP** **F** to reopen a completed flight in the wizard.

NOTE
Opening a saved flight or starting a new one closes the flight plan open in the wizard: save it first with **Save to Standby**.

CAUTION
Saved flights are stored in your account. On a shared device, select **Sign out** at the end of the session: it removes the local copy from the device. Graphic weather charts are not stored with the flight.

Figure 4.8 – Dashboard



A	Standby badge	Awaiting weather
B	Load briefing	Finalise
C	Edit data	Reopen
D	Ready to fly badge	Weather loaded
E	Update WX	LOAD newer briefing
F	View OFP	Reopen

SECTION 5

Operational Notes – Safety & Performance Limits

5.1 Aircraft applicability

WARNING

AFM data are those of the Reims FR172J (I-CCAF) and FR172H (I-CCAB); empty masses and moments are those of the two aircraft. With **Other**, speeds are those of the FR172J. Used for any other aircraft the results are not approximate: they are wrong.

5.2 Performance limits

ITEM	LIMITATION	EFFECT
Declared distances	TORA / TODA / LDA pre-filled with the physical runway length (OurAirports)	Must be corrected from the AIP where stopway, clearway or displaced threshold exist, or NOTAM changes apply
Tailwind	The AFM publishes no tailwind data	Distances are for zero wind and shorter than actual (red warning)
Wet / contaminated runway	No AFM data	No increment applied (red warning)
Grass runway	AFM grass increment	Applied when "Grass, dry" is selected
Landing mass	Landing table published at 1157 kg only	Conservative at lower masses
Touch and go	Computed with the take-off mass	Conservative
Headwind credit	Limited to the range of the AFM table / note	Grey advisory when limited
V-speeds	AFM table, no interpolation, conservative rounding	—
Weather for performance	Own METAR; nearest station within 30 NM on request; otherwise ISA standard values	Standard values flagged in red

5.3 Fuel planning as implemented

ITEM	METHOD
Trip	Climb from DEP elevation to cruise altitude (AFM fig. 5-6, 1157 kg, start-up and take-off included) + cruise over the remaining distance at the selected GS and fuel flow. Descent is not modelled separately (included in cruise). + aerodrome minutes : 5 min for each leg from or to an aerodrome (first and last leg; the two legs around an en-route aerodrome only if Touch & Go is selected), at cruise fuel flow.
Contingency	max(5% of Trip, 15 min at cruise fuel flow)
Alternate	Cruise, 22 inHg / 2400 RPM at the alternate altitude, over the alternate distance, + 5 aerodrome minutes. No climb allowance.
Final reserve	30 min (VFR) or 45 min (IFR) at cruise fuel flow
Taxi	(TO – OFF) + (ON – LDG) at 7.5 L/h
Units	1 L = 0.72 kg; 174 L (46 USG) usable

5.4 Navigation data

- Cruise GS is TAS (still air) until corrected for wind by the pilot.
- The alternate distance is a great-circle distance.

- Frequencies are the number only, from OurAirports: CROSS-CHECK with current charts / AIP.
- Navlog times and fuel are recomputed by the application; their totals equal the Trip.

5.5 Daylight check

- Sunrise / sunset: upper limb at -0.833° ; civil twilight: centre at -6° (night as defined in Regulation (EU) 923/2012).
- Rounded conservatively: sunset and end of twilight to the minute before, sunrise to the minute after.
- Uses the flight date of Step 8 and the UTC times of Step 1. No elevation correction (as in published tables).

5.6 Documents and storage

- The OFP is printed from the downloaded PDF, with the PDF viewer.
- Saved flights are stored in the pilot's account and are visible only to that pilot. On a shared device, **Sign out** at the end of the session.
- The A5 NAV-FLIGHTPLAN is a separate document and must be downloaded separately.

The documents produced are an aid to planning. Responsibility for the flight remains with the Pilot in Command.

APPENDIX A

Pre-Departure Cross-Check

Carry out before printing the documents. Each item refers to the section where the check is described.

ITEM	CHECK	REF.	✓
Times	OFF / TO in UTC; LDG, ON, BLOCK, FLT plausible	3.1	☐
Daylight	TO, LDG and ALTN landing reviewed; no unaccepted AFTER SUNSET / NIGHT	3.1	☐
Cruise	Altitude, RPM, MAP selected; GS corrected for wind	3.2	☐
Climb	Cruise altitude reached; TOC position plausible	3.2	☐
Alternate	ALTN entered; distance and altitude verified	3.2	☐
Fuel	Plan Block within 174 L usable; FOB not INSUFFICIENT	3.2	☐
Mass & balance	Masses within limits; CG within envelope at take-off and both landings	3.3	☐
Runways	Runway selected on every tab; TORA / TODA / LDA cross-checked with AIP and NOTAM	3.4	☐
Performance	All margins positive; every Checks & Warnings item reviewed	3.4	☐
Navlog	Totals equal Trip time and fuel; frequencies verified	3.5	☐
Weather & NOTAM	Correct aerodrome and current issue in every box; SIGMET / AIRMET box not empty	3.8	☐
Date / PIC	Flight date and PIC name entered	3.8	☐
Documents	OFP and NAV-FLIGHTPLAN downloaded / printed; OFP signed	4.2 / 4.3	☐